

ARTIFICIAL INTELLIGENCE IN FINTECH / WEEK 02

Financial Data and Returns

Version 2

Start with a stock-price record. Identify what it measures, calculate a return, and explain what that return leaves out.

Follow Microsoft data from prices to returns, then test the ideas against real company events and contrasting periods.

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Use the examples

Read the question, predict the named result, then inspect the code and output. The discussion questions are part of each issue; there is no separate practice section.

Run the code in order in a fresh Python notebook with numpy, pandas and yfinance installed. Outputs are shown here, so you can follow the calculations without running code during class.

Stock prices are Yahoo Finance observations retrieved through yfinance, not records of personal trades. Returns exclude trading fees and taxes. Dividends are included only where stated; no trading strategy is evaluated.

ISSUE 01

What data did we download?

First identify the company, field, currency and dates. MSFT is Microsoft common stock, listed on Nasdaq; its prices here are in USD per share.

The start date is included; the end date is excluded. A daily row labels a trading date, not a trade at midnight. This request uses daily data without pre-market or after-hours data.

Before running: Count the weekdays from August 14 through August 21. Will August 22 appear?

```
import numpy as np
import pandas as pd
import yfinance as yf

msft = yf.Ticker("MSFT").history(
    start="2024-08-14", end="2024-08-22",
    interval="1d", auto_adjust=False,
    actions=True, prepost=False, keepna=True,
)
msft.index = msft.index.strftime("%Y-%m-%d")
assert msft.index.is_unique
assert msft.index.is_monotonic_increasing
close = msft["Close"]
assert close.notna().all() and (close > 0).all()
print("Dates:", ", ", ".join(msft.index))
print("Rows:", len(msft))
```

Dates: 2024-08-14, 2024-08-15, 2024-08-16, 2024-08-19, 2024-08-20, 2024-08-21

Rows: 6

The request returns 6 rows, ending on August 21. August 17-18 are a weekend. Six closing prices give five adjacent close-to-close returns, because each return needs two prices.

Asset	Frequency	Currency
MSFT	Daily	USD

auto_adjust=False keeps Close and Adj Close separate. In the following tables, each row is a date, holding interval or event; each column is a variable or calculated result.

The checks stop calculation if dates repeat or prices are missing. They do not prove that the provider's observations are correct.

Discuss

1. Why can this request produce five returns, not six? Identify the extra price needed to calculate a return ending on August 14.
2. Read the request: what would changing end to August 21 remove, and which last return would disappear?

Sources: [1] yfinance: historical-data parameters; [2] Microsoft: dividend announced June 12, 2024. See the linked references at the end.

ISSUE 02

What do the price columns mean?

Open, High, Low and Close describe different prices within a trading session. Choose the field before choosing the formula.

Open and Close are the opening and closing prices; High and Low are the session extremes. Use O, H, L and C for those fields. Volume counts shares traded, not investors or money spent. Prices below are USD per share.

Before running: For each date, must both Open and Close lie between Low and High? Check both rows, not just one.

```
fields = ["Open", "High", "Low", "Close", "Volume"]
two_days = msft.loc[["2024-08-14", "2024-08-15"], fields]
result = two_days.copy()
for field in fields[:-1]:
    result[field + " (USD)"] = result.pop(field).map("{:.2f}".format)
result["Volume (shares)"] = result.pop("Volume").map("{:,}.0f".format)
result.insert(0, "Date", result.index)
result
```

Date	Open (USD)	High (USD)	Low (USD)	Close (USD)	Volume (shares)
2024-08-14	414.80	417.72	412.45	416.86	18,267,000
2024-08-15	419.80	421.11	417.66	421.03	20,752,100

MSFT's range was USD 3.45 per share; its open-to-close increase was USD 1.23 per share. The first measures the day's range, not a return earned by buying and selling.

Volume increased from 18,267,000 to 20,752,100 shares. That measures trading activity; it does not tell us how many different people traded.

The daily bar does not reveal whether the high came before the low. It also does not establish the exact price at which your order would have traded.

Discuss

1. Using these daily bars, can you claim that buying at the low and selling later at the high was possible? Name the missing information.
2. Compare the ranges and volumes on the two dates. Does greater volume necessarily imply a wider range in these two observations? Show both calculations.

Sources: [1] yfinance: historical-data parameters; [10] Yahoo Finance: MSFT historical data; [13] Nasdaq glossary: volume. See the linked references at the end.

ISSUE 03

How much did the closing price change?

A dollar gain and a percentage return answer different questions. Divide the price change by the starting price to express the change relative to the initial value.

For MSFT, the downloaded closes are USD 416.86 per share on August 14 and USD 421.03 on August 15. Subtract the first from the second, then divide by the first.

C_{t-1} and C_t are successive closing prices in USD per share. r_t is the simple price return, stored as a decimal. Both prices must use the same share basis, and the starting price must be positive.

$$r_t = \frac{C_t - C_{t-1}}{C_{t-1}} = \frac{C_t}{C_{t-1}} - 1$$

Before running: Will the dollar increase and percentage increase have the same numerical value? Explain which starting value the percentage uses.

```
start_close = close.loc["2024-08-14"]
end_close = close.loc["2024-08-15"]
price_change = end_close - start_close
price_return = price_change / start_close
print(f"Price change: USD {price_change:.2f} per share")
print(f"Price return: {price_return:.2%}")
```

Price change: USD 4.17 per share

Price return: 1.00%

The price increased by USD 4.17 per share, or 1.00%. A 1% return is not a USD 1 profit: it is a gain equal to 1% of the starting value.

This calculation covers the August 14 close to the August 15 close. It excludes the dividend, fees and taxes; it is not a record of a particular investor's realized profit.

Discuss

1. Convert the displayed percentage back to a decimal. Which starting value must you multiply it by to recover the price change, and why might rounded numbers differ slightly?
2. The result is positive. What information is still needed before saying that a student who traded MSFT on August 15 made this return?

Sources: [7] Zivot: Return Calculations; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 04

Can a stock rise after opening but still finish down?

A stock can finish above its opening price but below yesterday's close. The starting price determines which statement is true.

MSFT on August 21, 2024 is an example within our shared data. O is that day's open; C is that day's close; the preceding C is August 20's close. All three are USD per share, on the same share basis.

$$\frac{C_t}{C_{t-1}} = \frac{O_t}{C_{t-1}} \frac{C_t}{O_t}$$

Before running: Can the open-to-close return be positive while the full close-to-close return is negative?

```
prior = close.loc["2024-08-20"]
opening = msft.loc["2024-08-21", "Open"]
closing = close.loc["2024-08-21"]
overnight = opening / prior - 1
intraday = closing / opening - 1
full_day = closing / prior - 1
result = pd.DataFrame({
    "Date": ["2024-08-21"],
    "Overnight": [f"{overnight:.3%}"],
    "Open to close": [f"{intraday:.3%}"],
    "Close to close": [f"{full_day:.3%}"],
})
result
```

Date	Overnight	Open to close	Close to close
2024-08-21	-0.169%	0.014%	-0.155%

The opening price was already 0.169% below the prior close. The 0.014% rise after opening did not offset that decline: the close-to-close result remained -0.155%.

Multiplying the overnight and intraday growth factors recovers the full price ratio. The daily OHLC observations show these intervals; they do not prove that an investor executed orders at those exact prices.

Discuss

1. Reconcile the full-day return using $(1 + \text{overnight}) \times (1 + \text{intraday}) - 1$. Why should you multiply growth factors instead of comparing the signs alone?
2. Rewrite 'MSFT rose on August 21' so that the starting and ending prices are unambiguous. Then write the equally valid statement using the previous close.

Sources: [7] Zivot: Return Calculations; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 05

Does the return include the dividend?

A price-only return leaves out cash distributions. Microsoft's actual August 2024 dividend lets us add one item to the price calculation.

Stock	Dividend (USD/share)	Ex-date	Payment
MSFT	0.75	2024-08-15	2024-09-12

This calculation applies to a holder eligible for the announced dividend. The next issue explains how the ex-date determines that eligibility.

D_t is the dividend per share in USD. For a holder eligible for this dividend, value the amount owed at face value; exclude taxes, fees and reinvestment. The dividend is not cash paid on August 15.

$$r_t^{\text{div}} = \frac{C_t + D_t - C_{t-1}}{C_{t-1}}$$

Before running: Does adding the USD 0.75 dividend increase the return by 0.75%, or by the dividend divided by the starting price?

```
dividend = msft.loc["2024-08-15", "Dividends"]
dividend_return = (end_close + dividend) / start_close - 1
result = pd.DataFrame({
    "Period (2024)": ["Aug 14-15"],
    "Price only": [f"{price_return:.2%}"],
    "Price + dividend": [f"{dividend_return:.2%}"],
})
result
```

Period (2024)	Price only	Price + dividend
Aug 14-15	1.00%	1.18%

Including the dividend changes the result from 1.00% to 1.18%. Adj Close also incorporates applicable distributions through price adjustments: do not add the dividend to its return again. Its ratio need not exactly equal this cash-dividend calculation.

Discuss

1. Which event date determines whether this dividend belongs in the holding-period calculation? Explain why the September payment date is not enough.
2. For August 14-15, choose Close or Adj Close for a price-only calculation. What is wrong with using an adjusted-price return and then adding the same dividend again?

Sources: [2] Microsoft: dividend announced June 12, 2024; [3] Investor.gov: ex-dividend dates; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 06

Can you sell on the ex-date and keep the dividend?

The ex-dividend date separates shares carrying the upcoming dividend entitlement from shares trading without it. The payment date serves a different purpose.

Microsoft's actual announcement sets August 15, 2024 as the ex-date and September 12 as the payment date, for USD 0.75 per share. Apply the regular cash-dividend rule to these dates; the table describes eligibility, not observed investor accounts.

Date	Action	Upcoming dividend
2024-08-14	Buy before the ex-date	Buyer normally eligible
2024-08-15	Buy on the ex-date	Buyer not eligible
2024-08-15	Sell an eligible holding	Seller keeps entitlement
2024-09-12	Buy on the payment date	Buyer not eligible

Why selling on August 15 does not remove the entitlement

On the ex-date, the buyer purchases the stock without this upcoming dividend. An already eligible seller retains the entitlement, even though the cash arrives later. Buying before the September payment does not restore an entitlement that the purchase missed.

The return calculation and the eligibility question are separate. First establish entitlement; only then decide whether to include the USD 0.75 dividend. These rules concern this regular cash dividend, not every special dividend or stock distribution.

Discuss

1. Using the announced dates, explain why selling an eligible holding on August 15 and buying on August 15 lead to different dividend rights.
2. Does the September 12 payment date require an already eligible holder to retain the stock until then? Explain what the ex-date determines and what the payment date determines.

Sources: [2] Microsoft: dividend announced June 12, 2024; [3] Investor.gov: ex-dividend dates. See the linked references at the end.

ISSUE 07

Does a stock split make an investor richer?

A split changes the number of shares and the price per share. It does not, by itself, create investment value.

Apple announced a four-for-one split in 2020. Split-adjusted trading began on August 31. The announcement grants three additional shares for each existing share; this is an actual corporate action, not an invented portfolio.

Event	Share count	Price per share	Position value
AAPL 2020 split	Multiply by 4	Divide by 4	Unchanged

Let N be a share count and C a price in USD per share on the pre-split basis. Holding all other price changes aside, the same ownership value is represented after the split as follows.

$$(4N)\left(\frac{C}{4}\right) = NC$$

Before running: The downloaded history uses a common split-adjusted share basis. Should its August 28-31 price return mechanically be -75%?

```
apple = yf.Ticker("AAPL").history(
    start="2020-08-28", end="2020-09-01",
    auto_adjust=False, actions=True,
)
split = apple.loc["2020-08-31", "Stock Splits"]
split_day_return = apple["Close"].iloc[-1] / apple["Close"].iloc[0] - 1
print(f"Recorded split: {split:.0f}-for-1")
print(f"Close-series return, Aug 28-31: {split_day_return:.2%}")
```

Recorded split: 4-for-1
Close-series return, Aug 28-31: 3.39%

The split record is 4-for-1, while the downloaded Close series rises 3.39%. The split identity does not predict the actual market return. Do not divide this already split-adjusted history by four again.

Discuss

1. Why does a fourfold increase in shares not imply a fourfold increase in value? Use the split identity, not an assumed account balance.
2. Why would applying the split ratio again corrupt this return? Identify which input's share basis you would check before changing a price history.

Sources: [4] Apple: four-for-one split announcement; [1] yfinance: historical-data parameters; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 08

How much is the stock worth in TWD?

To express a USD price in New Taiwan dollars, multiply by the number of TWD per USD. Keep the quotation direction visible.

The Federal Reserve reports TWD 32.27 per USD for August 15, 2024. This is a New York noon reference observation, not an exchange rate synchronized with the stock close. Use it here only to translate that day's closing value, not to claim an executable trade.

C_t is the USD closing price per share. X_t is the reference rate in TWD per USD. V_t is the translated value in TWD per share.

$$V_t = C_t X_t$$

The USD units cancel: (USD per share) multiplied by (TWD per USD) gives TWD per share.

Before running: Will the numerical TWD value be larger or smaller than the USD price? Does that numerical change mean an investment gain?

```
# Federal Reserve H.10: observed Aug 15, released Aug 19, 2024.
twd_per_usd = 32.27
twd_value = end_close * twd_per_usd
print(f"Reference rate: {twd_per_usd:.2f} TWD per USD")
print(f"Translated close: TWD {twd_value:,.2f} per share")
```

```
Reference rate: 32.27 TWD per USD
Translated close: TWD 13,586.64 per share
```

The translated value is TWD 13,586.64 per share. It expresses the same stock-price observation in another currency; it does not show a new profit.

The H.10 page was released on August 19. It is suitable for this retrospective calculation, but that publication cannot be treated as information already available on August 15.

Discuss

1. Why is multiplication appropriate for a rate labeled TWD per USD? Show the cancellation of units.
2. Can this conversion establish what a broker would have paid in TWD at the August 15 close? Identify the observation-time and transaction information that is missing.

Sources: [5] Federal Reserve: H.10, August 19, 2024; [6] FRED: DEXTAUS units and observation time. See the linked references at the end.

ISSUE 09

Can a USD gain become a TWD loss?

A rising USD share price does not guarantee a rising TWD value. Each dollar can buy fewer TWD at the end of the same period.

Reuse MSFT closes on August 15 and 19. Federal Reserve reference rates are 32.27 and 32.01 TWD per USD on those dates. As before, these noon rates translate closing values retrospectively; they are not synchronized closing FX quotes or realized exchange proceeds.

R_{USD} is the stock's price return; R_{FX} is the change in TWD per USD; R_{TWD} is the return on the translated value. Subscripts 0 and 1 mean August 15 and 19. All returns are decimals in the formula.

$$1 + R_{TWD} = \frac{C_1 X_1}{C_0 X_0} = (1 + R_{USD})(1 + R_{FX})$$

Before running: The stock price rises slightly while TWD per USD falls. Which change is larger, and can the translated value decline?

```
fx_dates = ["2024-08-15", "2024-08-19"]
# Actual H.10 observations from the two cited weekly releases.
fx_rates = pd.Series([32.27, 32.01], index=fx_dates)
usd_prices = close.loc[fx_dates]
translated = usd_prices * fx_rates
usd_return = usd_prices.iloc[-1] / usd_prices.iloc[0] - 1
fx_return = fx_rates.iloc[-1] / fx_rates.iloc[0] - 1
twd_return = translated.iloc[-1] / translated.iloc[0] - 1
result = pd.DataFrame({
    "Period (2024)": ["Aug 15-19"],
    "USD return": [f"{usd_return:.3%}"],
    "FX change": [f"{fx_return:.3%}"],
    "TWD return": [f"{twd_return:.3%}"],
})
result
```

Period (2024)	USD return	FX change	TWD return
Aug 15-19	0.119%	-0.806%	-0.688%

MSFT's USD price rose 0.119%, but each USD bought 0.806% fewer TWD. The translated value fell 0.688%. The currency movement outweighed the stock-price increase.

No dividend is included: this interval starts at the August 15 close, on the ex-date. Fees and taxes are excluded. The reference-rate releases occurred on August 19 and 26; these calculations are not a historical trading signal.

Discuss

1. Use the two growth factors to reproduce the TWD result. Which component reverses the sign of the stock-price gain?
2. What additional FX observations and costs would be needed to describe an executable TWD investment return rather than this reference-rate translation?

Sources: [5] Federal Reserve: H.10, August 19, 2024; [6] FRED: DEXTAUS units and observation time; [11] Federal Reserve: H.10, August 26, 2024. See the linked references at the end.

ISSUE 10

Why use log returns?

Simple returns describe proportional gains. Log returns describe the same price ratios in a form that can be added over consecutive holding intervals.

For positive closing prices, let r_t be the simple return and ℓ_t the log return. $\ln$ is the natural logarithm; $\exp$ reverses it. Both are stored as decimals, not dollar amounts.

$$\ell_t = \ln\left(\frac{C_t}{C_{t-1}}\right) = \ln(1 + r_t), \quad r_t = e^{\ell_t} - 1$$

Before running: For the positive August 14-15 price return, will the log return be slightly smaller than the simple return?

```
log_return = np.log(end_close / start_close)
result = pd.DataFrame({
    "Period (2024)": ["Aug 14-15"],
    "Simple return": [f"{price_return:.4%}"],
    "Log return": [f"{log_return:.4%}"],
})
result
```

Period (2024)	Simple return	Log return
Aug 14-15	1.0003%	0.9954%

The simple return is 1.0003%; the log return is 0.9954%. They describe the same price ratio. Their difference is not a fee or a loss. Convert the log return back before calling it a percentage gain.

Why researchers use it

The logarithm turns a product of consecutive growth factors into a sum. Researchers can add daily log returns to form weekly or monthly log returns, and express return as a change in log price. This is useful algebra; it does not make returns independent or normally distributed.

Discuss

1. Recover the simple return with $\exp(\log_return) - 1$. Why does subtracting the two displayed returns fail to measure any money lost?
2. To summarize this price movement as a percentage gain, which value would you report? Explain one different task for which the log value is convenient.

Sources: [7] Zivot: Return Calculations. See the linked references at the end.

ISSUE 11

What is the return over several days?

For the full period, compare the last close with the first close. Daily percentage changes must compound because each starts from a different price.

Use the same six MSFT closes, August 14-21. Let R be the full-period price return. The first two growth factors illustrate why intermediate prices cancel.

$$(1 + r_1)(1 + r_2) = \frac{C_1}{C_0} \frac{C_2}{C_1} = \frac{C_2}{C_0}$$

Before running: On August 16, the daily price return is negative. Does that necessarily make the return since August 14 negative?

```
daily_return = close.pct_change(fill_method=None).iloc[1:]
cumulative_return = (1 + daily_return).cumprod() - 1
endpoint_return = close.iloc[-1] / close.iloc[0] - 1
log_sum = np.log1p(daily_return).sum()
result = pd.DataFrame({
    "Date": daily_return.index,
    "Daily return": daily_return.map(lambda r: f"{r:.2%}"),
    "Since Aug 14": cumulative_return.map(lambda r: f"{r:.2%}"),
})
result
```

Date	Daily return	Since Aug 14
2024-08-15	1.00%	1.00%
2024-08-16	-0.61%	0.39%
2024-08-19	0.73%	1.12%
2024-08-20	0.78%	1.90%
2024-08-21	-0.16%	1.75%

August 16's daily return is -0.61%, but the gain since August 14 remains 0.39%. The last cumulative value is 1.75%.

Using only the first and last close gives 1.7464%, equal to the compounded result before rounding. Converting the sum of log returns with $\exp(\log_sum) - 1$ also gives 1.7464%. These are three ways to describe the same price ratio, not three different investment outcomes.

Discuss

1. Use just August 15 and 16's daily returns to recover the return since August 14 at August 16. Explain why adding those two percentages is not the exact calculation.
2. If one day's return is omitted, why is equality with the endpoint return no longer assured? Identify the missing price ratio.

Sources: [7] Zivot: Return Calculations; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 12

Which average return should we use?

Use the arithmetic mean to describe the average observed one-period return. Use the geometric mean to find a constant per-period return that compounds to the same ending value.

There are $n = 5$ close-to-close returns in the same August 14-21 sample. The bar over r denotes their arithmetic mean; g is their geometric mean. R is the full-period return from Issue 11. All rates are decimals in the formulas.

$$\bar{r} = \frac{r_1 + \dots + r_n}{n}, \quad (1 + g)^n = 1 + R$$

To find g , take the n th root of the full-period growth factor and subtract one. This asks for an equivalent constant rate, not a forecast of tomorrow's return.

$$g = (1 + R)^{1/n} - 1$$

Before running: Which of these averages, repeated for five intervals, must reproduce the observed full-period return?

```
n = len(daily_return)
arithmetic_mean = daily_return.mean()
geometric_mean = (1 + endpoint_return)**(1 / n) - 1
result = pd.DataFrame({
    "Period (2024)": ["Aug 14-21"],
    "Arithmetic mean": [f"{arithmetic_mean:.4%}"],
    "Geometric mean": [f"{geometric_mean:.4%}"],
})
result
```

Period (2024)	Arithmetic mean	Geometric mean
Aug 14-21	0.3488%	0.3469%

The averages are 0.3488% and 0.3469% per close-to-close interval. Compounding the geometric mean five times gives 1.7464%, the observed full-period price return.

Here 'per interval' means one adjacent pair of trading-date closes. A Friday-to-Monday interval includes the weekend; it is not one calendar day.

Discuss

1. Calculate $(1 + \text{geometric_mean})^5 - 1$ and compare it with Issue 11. What exactly does their agreement establish?
2. A report calls the arithmetic mean the 'constant daily growth rate.' Rewrite that statement using the appropriate average, without claiming a future return.

Sources: [7] Zivot: Return Calculations. See the linked references at the end.

ISSUE 13

Can a positive average end in a loss?

A positive arithmetic mean does not guarantee a positive compounded return. MSFT on March 6-10, 2020 provides a real counterexample.

This period is selected to demonstrate the distinction, not to estimate how often it occurs. Use the closes on March 6, 9 and 10: three prices produce two close-to-close returns.

For two simple returns r_1 and r_2 , the cumulative return R includes their product as well as their sum. Opposite signs make that product negative.

$$R = (1 + r_1)(1 + r_2) - 1 = r_1 + r_2 + r_1r_2$$

Before running: The second return is positive. Does adding the two returns establish that the ending close exceeds the first close?

```
march = yf.Ticker("MSFT").history(
    start="2020-03-06", end="2020-03-11", auto_adjust=False,
)
march_close = march["Close"]
march_returns = march_close.pct_change(fill_method=None).iloc[1:]
march_mean = march_returns.mean()
march_total = march_close.iloc[-1] / march_close.iloc[0] - 1
result = pd.DataFrame({
    "Return ending": march_returns.index.strftime("%Y-%m-%d"),
    "Simple return": march_returns.map(lambda r: f"{r:.4%}"),
})
result
```

Return ending	Simple return
2020-03-09	-6.7773%
2020-03-10	6.8384%

The two returns are -6.7773% and 6.8384%. Their arithmetic mean is positive, 0.0306% per interval, yet the full-period price return is -0.4023%.

The rebound applies to the lower value left after the loss. A percentage rise measured from that lower base can exceed the earlier percentage decline without fully recovering the starting value. The product term shows exactly why simply adding rates fails.

Discuss

1. Calculate $r_1 \cdot r_2$ and add it to $r_1 + r_2$. Which term changes the sign of the full-period result in this example?
2. After the March 9 decline, calculate the return needed to recover the March 6 close. Compare it with the actual March 10 return; do not round before comparing.

Sources: [7] Zivot: Return Calculations; [10] Yahoo Finance: MSFT historical data. See the linked references at the end.

ISSUE 14

Is an absent date a missing price?

No row on a non-trading day is different from a missing price on a trading day. Do not turn either situation into a measured zero return.

MSFT's August 16 and 19 observations are Friday and Monday. August 17-18 have no regular-session daily bars. Nasdaq's regular session was 09:30-16:00 New York time on these trading dates; detailed time-zone conversion is not needed for this calculation.

Before running: Which two calendar dates should have no regular-session row? Does their absence mean the download lost a trading day's price?

```
result = pd.DataFrame({
    "Date": ["2024-08-16", "2024-08-17", "2024-08-18", "2024-08-19"],
    "Day": ["Friday", "Saturday", "Sunday", "Monday"],
})
result["Row returned"] = result["Date"].isin(close.index).map({True: "Yes", False: "No"})
result
```

Date	Day	Row returned
2024-08-16	Friday	Yes
2024-08-17	Saturday	No
2024-08-18	Sunday	No
2024-08-19	Monday	Yes

The Saturday and Sunday rows are absent from the source, not deleted for this example. The return ending on Monday uses Friday's close. Its interval spans three calendar days, but only one adjacent close-to-close step.

Before running: Can all six prices be present while the first return is still missing?

```
returns_with_first = close.pct_change(fill_method=None)
print("Missing Close values:", int(close.isna().sum()))
print("Missing return values:", int(returns_with_first.isna().sum()))
print("First return date:", returns_with_first.index[0])
```

```
Missing Close values: 0
Missing return values: 1
First return date: 2024-08-14
```

There are no missing Close values and one missing return. The return ending August 14 has no starting close inside this sample. It is unknown, not zero; use an earlier observation if you need that return.

Discuss

1. Before filling a gap on a weekday, what would you check in the exchange calendar and source request? Why is a weekday alone not proof that the market was open?
2. Why does filling the first missing return with zero change the average's meaning? State which missing starting observation would resolve the problem instead.

Sources: [1] yfinance: historical-data parameters; [8] Nasdaq: market hours and calendar. See the linked references at the end.

ISSUE 15

Should we delete a very large return?

A large price change deserves investigation, not automatic deletion. Netflix provides a real event that the small Microsoft sample does not contain.

Netflix released its Q1 2022 shareholder letter on April 19. Calculate the closing-price change from April 19 to April 20; then separate the observed change from an explanation of its cause.

Before running: Would a rule that deletes every daily return below -20% remove this observation? Would that rule, by itself, establish that the price is wrong?

```
netflix = yf.Ticker("NFLX").history(
    start="2022-04-19", end="2022-04-21",
    auto_adjust=False, actions=True, keepna=True,
)
nflx_close = netflix["Close"]
assert len(nflx_close) == 2 and nflx_close.notna().all()
nflx_return = nflx_close.iloc[-1] / nflx_close.iloc[0] - 1
print(f"NFLX price return, Apr 19-20: {nflx_return:.2%}")
print("Split events in request:", int(netflix["Stock Splits"].ne(0).sum()))
```

```
NFLX price return, Apr 19-20: -35.12%
Split events in request: 0
```

The source gives a -35.12% price return and no split event in the request. The shareholder letter documents a real company announcement. It does not establish that one news item explains the entire price change, or independently verify both price observations.

Before correcting a value

Confirm the security, dates, price field and share basis; compare the prices with another reliable source and inspect company events. Correct a documented error, not an uncomfortable result. A rule can flag an observation for review without removing it.

Discuss

1. Would you delete this row solely because its return is below -20%? Identify the evidence needed to justify changing the recorded price.
2. Why can deleting this loss make a return summary misleading? What does the dated shareholder letter support, and what causal claim does it not establish?

Sources: [1] yfinance: historical-data parameters; [9] Netflix: Q1 2022 shareholder letter. See the linked references at the end.

ISSUE 16

Does a new ticker mean a new stock?

Before joining price histories, establish whether a ticker change represents the same security. A different label does not necessarily mean a different investment.

Meta's May 31, 2022 SEC-filed announcement states that its Class A common stock would switch from FB to META before the June 9 market open. Nasdaq listing and the security's CUSIP identifier would remain unchanged.

Date	Ticker	Company	Share class	Market
2022-06-08	FB	Meta Platforms	Class A	Nasdaq
2022-06-09	META	Meta Platforms	Class A	Nasdaq

Read the record before joining prices

These are two dated labels for the same Class A security, based on the announcement. Shareholders did not need to exchange their shares merely because the ticker changed. The announcement supplies identity evidence, not a new price or a guaranteed return.

A provider may show today's ticker on the entire historical series. That can be convenient for retrieval, but it does not establish which ticker was used on a past trading date. Check the provider's history rather than treating a failed FB request as proof that earlier prices do not exist.

Discuss

1. Which facts in the announcement support joining the FB and META histories? Explain why ticker text alone would miss this connection.
2. What checks are still necessary before computing the June 8-9 return from two files? State the dates, price field, units and share-basis conditions; the announcement alone does not supply them.

Sources: [12] Meta: ticker change, SEC exhibit dated May 31, 2022. See the linked references at the end.

REFERENCES & REPRODUCTION

Sources and calculation notes

Tables show selected observations, event facts and calculated results. The two-day OHLCV excerpt supports field interpretation; full downloaded files are not included. FX observations are transcribed from the Federal Reserve's dated releases. Rows identify samples or events; columns identify variables.

- [1] [yfinance: historical-data parameters](#)
- [2] [Microsoft: dividend announced June 12, 2024](#)
- [3] [Investor.gov: ex-dividend dates](#)
- [4] [Apple: four-for-one split announcement](#)
- [5] [Federal Reserve: H.10, August 19, 2024](#)
- [6] [FRED: DEXTAUS units and observation time](#)
- [7] [Zivot: Return Calculations](#)
- [8] [Nasdaq: market hours and calendar](#)
- [9] [Netflix: Q1 2022 shareholder letter](#)
- [10] [Yahoo Finance: MSFT historical data](#)
- [11] [Federal Reserve: H.10, August 26, 2024](#)
- [12] [Meta: ticker change, SEC exhibit dated May 31, 2022](#)
- [13] [Nasdaq glossary: volume](#)

Reproduce the examples

Run Issue 1 first, then continue in order. Each last table expression displays in a Python notebook; in a plain Python script, use `print(result)`. Install dependencies only if they are missing: `python -m pip install numpy pandas yfinance`.

Executed 2026-09-14 (UTC). Python 3.12.9; numpy 2.2.6; pandas 2.2.3; yfinance 1.2.0. The printed summaries and tables were generated by those displayed code blocks. Full-precision values are used before rounding for display.

Historical provider observations can be revised. Internet access and provider availability are required to repeat downloads. Yahoo's historical-page web preview was unavailable during this build; downloads through yfinance succeeded. Price-field conventions follow the course's prior source check; the exact adjustment algorithm is not re-derived here.

This short sample teaches calculation, not expected returns or a trading recommendation. Reading the log-return identity does not establish a distributional model. Calendar checks do not establish that every observed price is correct.